Measure sound speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Analyze sound travel measurements and their limitations.

SUCCESS CRITERIA - I can...

1. Define one-way versus round-trip paths.
2. Calculate speed from distance and time.
3. Use repeated measurements and discuss timing uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “trial” and “echo” to explain your model or result. Sentence starters are optional.

trial: one run or repetition of a test

echo: a sound received after reflection

round trip: travel from start to a reflector and back

01 Bring back an earlier idea

Three instantaneous displacements are +3 mm, -5 mm and -7 mm.
Find their sum and explain why one displacement value does not determine total wave energy.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: v = d/Δt; v = 2 d/Δt for an echo.
Match the distance to one-way or round-trip travel. Use moderate sound levels and record actual data separately.

Worked reasoning: A sound echo returns 0.4 s after a pulse. At 340 m/s, round-trip distance is 136 m, so the reflector is 68 m away.

Your turn: Draw outgoing and returning paths and mark where the factor of 2 enters.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

A pulse travels between two sensors separated by 1.70 m. Delay is .0050 s. Is this one-way or echo distance?
Find speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Solve and justify

An echo returns after .0200 s from a wall 3.40 m away.
Find speed. If delay uncertainty is±.0005 s with distance exact, find speed bounds.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Plan a table for three trials. Include distance, elapsed time, method (one-way or echo), speed and one possible systematic error.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

06 Choose a method

A sphere’s radius becomes 4 times as large at fixed density. Derive the area-to-mass multiplier.
Explain why kinetic energy per mass alone does not predict survival during entry.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

07 Exit check

Two sensors are .85 m apart and record a delay .0025 s. **Find** v. A constant .001 s timing offset is added; find the incorrect v and describe bias.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A microphone pair separated 1.5 m measures travel-time delay 0.0044 s.
Find speed. If delay is uncertain by±0.0002 s, find worst-case speed bounds.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3